Measure Theory

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Problem

Consider a random (infinite) string of 0's and 1's

$$(\ell_1, \ldots, \ell_n, \ldots) \quad \ell_n \in \{0,1\} \forall n \in \mathbb{N}$$

What is the probability that somewhere along the string, I find 100 consecutive zeros?

Comment

The statement of the problem is unclear. Lebesgue's machinery will clarify this

Define

$$X = \{x = (\ell_1, \ldots, \ell_n, \ldots) \mid \ell_n \in \{0,1\} \forall n \in \mathbb{N}\}$$

This is, in a natural way, a compact metric space. Use the "Hamming-style" distance:
For $x = (\ell_1, \ldots, \ell_n, \ldots)$ and $x' = (\ell_1{}', \ldots, \ell_n{}', \ldots)$

$$d(x, x') = \sum_{n \in \mathbb{N}} \frac{|\ell_n - \ell_n{}'|}{2^n}$$

Since this is a metric space, we can consider the topology:

$$\mathcal{T} = \{G \subseteq X \mid G \text{ is open}\} \subseteq 2^X = \mathcal{P}(X)$$

Can consider

$$\mathfrak{B} = \sigma\text{-Alg}(\mathcal{T}) \quad \text{(Borel sigma algebra)}$$

Then a probability measure is

$$P : \mathcal{B} \longrightarrow [0,1]$$

Problem 1.1 says:
Let

$$S = \left\{x = (\ell_1, \ldots, \ell_n, \ldots) \in X \;\middle|\; \exists m \in \mathbb{N} \text{ with } \ell_m = \ell_{m+1} = \cdots = \ell_{m+99} = 0\right\}$$

check that $S \in \mathcal{B}$, and compute $P(S)$

1 Algebras of Sets

Fix a non-empty set X.

Definition 1.1 **Algebra of Sets**

$\mathfrak{A} \subseteq 2^X$ is an *algebra of sets* if

1. $\varnothing \in \mathfrak{A}$
2. If $A \in \mathfrak{A}$, then $X \setminus A \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \cap B \in \mathfrak{A}$

Proposition 1.2 **Properties of algebras of sets**

Let $\mathfrak{A}$ be an algebra of sets on X. Then

1. $X \in \mathfrak{A}$
2. If $A, B \in \mathfrak{A}$, then $A \cup B \in \mathfrak{A}$
3. If $A, B \in \mathfrak{A}$, then $A \setminus B \in \mathfrak{A}$
4. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcap \mathfrak{B} \in \mathfrak{A}$
5. If $\mathfrak{B} \subseteq \mathfrak{A}$ is finite, then $\bigcup \mathfrak{B} \in \mathfrak{A}$

Example 1.1

Obviously, 2^X and $\{\varnothing, X\}$ are algebras of sets on X

Lemma 1.3 **Intersections of algebras**

Let $\mathfrak{F}$ be any collection of algebras of sets on X, then $\bigcap \mathfrak{F}$ is an algebra of sets on X.

Proof: First note if $\mathfrak{F} = \varnothing$, then $\bigcap \mathfrak{F} = 2^X$ which is an algebra of sets on X. We now proceed with checking the axioms of an algebra of sets on X.

1. For every $\mathfrak{A} \in \mathfrak{F}$, we have $\varnothing \in \mathfrak{A}$, thus $\varnothing \in \bigcap \mathfrak{F}$.
2. If $A \in \bigcap \mathfrak{F}$, then $A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $X \setminus A \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $X \setminus A \in \bigcap \mathfrak{F}$.
3. If $A, B \in \bigcap \mathfrak{F}$, then $A, B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$. Thus $A \cap B \in \mathfrak{A}$ for every $\mathfrak{A} \in \mathfrak{F}$, and thus $A \cap B \in \bigcap \mathfrak{F}$.

$\square$

Proposition 1.4 **Algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X, then there exists a unique algebra of sets $\mathfrak{A}$ on X with the following properties:

1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is an algebra of sets on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: Define $\mathfrak{F}$ to be the collection of all algebras of sets on X that contain $\mathfrak{B}$. Then $\bigcap \mathfrak{F}$ is an algebra of sets by Lemma 1.3 and satisfies (1) and (2). Uniqueness follows from (2). $\square$

Definition 1.5 **Algebra of sets generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X. We define the *algebra of sets generated by* $\mathfrak{B}$ to be the unique algebra from Proposition 1.4. We notate it as $\mathrm{Alg}(\mathfrak{B})$.

1.1 Additive set-function on an algebra of sets

Definition 1.6 **Additive set-function**

Let $\mathfrak{A} \subseteq 2^X$ be an algebra of sets on X. A function $\mu : \mathfrak{A} \to [0, \infty]$ is called an *additive set-function on* $\mathfrak{A}$ if

1. If $A, B \in \mathfrak{A}$ with $A \cap B = \varnothing$, then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. $\mu(\varnothing) = 0$

Remark 1.7

The arithmetic of $[0, \infty]$ is standard, except for $0 \cdot \infty = \infty \cdot 0 = 0$, which is a useful convention in measure theory.

Remark 1.8

We almost don't need to require (2), it's just when μ is identically ∞ do things go wrong without (2)

Proposition 1.9 **Properties of an additive set function**

Let $\mathfrak{A}$ be an algebra of sets on X, and let μ be an additive set function on $\mathfrak{A}$. Then

1. If $A, C \in \mathfrak{A}$ with $A \subseteq C$, then $\mu(A) \leq \mu(C)$
2. If $A, B \in \mathfrak{A}$, then $\mu(A \cup B) \leq \mu(A) + \mu(B)$ (sub additivity)
3. If $A_1, \ldots, A_n \in \mathfrak{A}$ are all pairwise disjoint, then $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$
4. If $A_1, \ldots, A_n \in \mathfrak{A}$, then $\mu(A_1 \cup \cdots \cup A_n) \leq \mu(A_1) + \cdots + \mu(A_n)$

2 Sigma Algebras

Definition 2.1 **Sigma Algebra**

$\mathfrak{M} \subseteq 2^X$ is called a *sigma algebra* on X if

1. $\varnothing \in \mathfrak{M}$
2. If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$
3. If $(A_n)_{n=1}^{\infty} \subseteq \mathfrak{M}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathfrak{M}$

Remark 2.2

countable unions work too, and so do finite operations

Lemma 2.3 **Intersections of sigma algebras**

Let $\mathfrak{F}$ be any collection of sigma algebras on X, then $\bigcap \mathfrak{F}$ is a sigma algebra on X.

Proof: basically identical to the proof for algebras of sets $\square$

Proposition 2.4 **Sigma algebra generated by a collection**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X, then there exists a unique sigma algebra $\mathfrak{A}$ on X with the following properties:

1. $\mathfrak{B} \subseteq \mathfrak{A}$
2. Whenever $\mathfrak{C}$ is a sigma algebra on X with $\mathfrak{B} \subseteq \mathfrak{C}$, then we have $\mathfrak{A} \subseteq \mathfrak{C}$

Proof: basically identical to the proof for algebras of sets □

Definition 2.5 **Sigma algebra generated by a collection of subsets**

Let $\mathfrak{B} \subseteq 2^X$ be any collection of subsets of X. We define the *sigma algebra generated by* $\mathfrak{B}$ to be the unique sigma algebra from Proposition 2.4. We notate it as $\sigma\text{-Alg}(\mathfrak{B})$.

Definition 2.6 **Borel Sigma Algebra**

Given a topological space $(X, \mathcal{T})$, the sigma algebra generated by $\mathcal{T}$ is called the *Borel sigma algebra on* X and written $\mathcal{B}_X := \sigma\text{-Alg}(\mathcal{T})$

2.1 Positive measure on a sigma algebra

Remark 2.7 **Increasing limits and series in $[0, \infty]$**

1. Let $(s_n)_{n=1}^{\infty}$ be an *increasing* sequence in $[0, \infty]$ in the sense that

$$s_1 \leq s_2 \leq \cdots$$

Such a sequence is sure to approach a limit $L \in [0, \infty]$. With a slight abuse of notation we can also write the formula $L = \sup\{s_n \mid n \in \mathbb{N}\}$ meaning that L is the smallest upper bound for all the s_n's. This is usually written in connection to a bounded increasing sequence of numbers in $[0, \infty)$ – our "slight abuse of notation" is that we are also allowing it to cover the case $L = \infty$, which can occur when some of the s_n's are equal to ∞, or when $(s_n)_{n=1}^{\infty}$ is an unbounded sequence in $[0, \infty)$.

2. Let $(t_n)_{n=1}^{\infty}$ be any sequence in $[0, \infty]$. We can consider the finite sums

$$s_n := \sum_{i=1}^{n} t_i$$

which forms an increasing sequence in $[0, \infty]$. We can consider $L = \lim_{n \to \infty} s_n$ as in (1.), and use the familiar notation $L = \sum_{n=1}^{\infty} t_n$

Definition 2.8 **Positive measure**

Let $\mathfrak{M} \subseteq 2^X$ be sigma algebra of subsets of X. We say $\mu : \mathfrak{M} \to [0, \infty]$ is a *positive measure* on $\mathfrak{M}$ if it satisfies:

1. $\mu(\varnothing) = 0$
2. $$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

   Whenever $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$ with the property that if $i, j \in \mathbb{N}$ with $i \neq j$ we have $A_i \cap A_j = \varnothing$

Remark 2.9

A positive measure is, in particular, finitely additive. Thus we can use all the properties in Proposition 1.9

Definition 2.10 **Nomenclature involving measures**

1. If $\mathfrak{M}$ is a sigma algebra on X, then the pair $(X, \mathfrak{M})$ is called a *measurable space*
2. If we also have a positive measure μ on $\mathfrak{M}$, then the triple $(X, \mathfrak{M}, \mu)$ is called a *measure space*
3. If $\mu(X) < \infty$, then we say μ is a *finite measure*
4. If $\mu(X) = 1$, then we say μ is a *probability measure* and that $(X, \mathfrak{M}, \mu)$ is a *probability space*
5. We say that μ is *sigma finite* if there exists a sequence of sets $(X_n)_{n=1}^{\infty}$ in $\mathfrak{M}$ with
   a. $X_1 \subseteq X_2 \subseteq \cdots$ (The X_n's are said to form an increasing chain)
   b. $\bigcup_{n=1}^{\infty} X_n = X$
   c. $\mu(X_n) < \infty$ for all $n \in \mathbb{N}$

Proposition 2.11 **Sigma sub-additivity**

Let $\mathfrak{M}$ be a sigma algebra on X, and let μ be a positive measure on $\mathfrak{M}$. If $(A_n)_{n=1}^{\infty}$ is a sequence of sets in $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Proposition 2.12 **Continuity along chains**

Let $\mathfrak{M}$ be a sigma algebra on X, and let μ be a positive measure on $\mathfrak{M}$.

1. If $(A_n)_{n=1}^{\infty}$ is an increasing chain in $\mathfrak{M}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

2. If $(A_n)_{n=1}^{\infty}$ is a decreasing chain in $\mathfrak{M}$ with $\mu(A_1) < \infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n)$$

Example 2.1

Example of "atomic" μ based on weights of individual points $x \in X$. Suppose you have $w : X \to [0, \infty]$. Let $\mathfrak{M} = 2^X$, for every $A \subseteq X$, put

$$\mu(A) = \sup\left\{\sum_{x\in F} w(x) \;\middle|\; F \subseteq A \text{ finite}\right\}$$

Then $(X, \mathfrak{M}, \mu)$ is a measure space. For the special case that $w(x) = 1, \forall x \in X$ is called the *counting measure* on X. For the special case that $w(x_0) = 1$ for some $x_0 \in X$ and $w(x) = 0, \forall x \neq x_0$ is called the *Dirac measure concentrated at* x_0

2.2 A glimpse of the Lebesgue measure on $\mathbb{R}$

Note

1. $\mathcal{B}_{\mathbb{R}}$ can be generated from the half open intervals
2. If $\mu, \nu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ are positive measures such that

$$\mu((a, b]) = \nu((a, b]) < \infty, \forall a < b \in \mathbb{R}$$

   It follows that $\mu = \nu$

Then the *Lebesgue measure* $\lambda : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ is determined via with requirement that $\lambda((a, b]) = b - a, \forall a < b \in \mathbb{R}$.
Another approach: λ is uniquely determined by the requirement that $\lambda(B + t) = \lambda(B)$ for every $B \in \mathcal{B}_{\mathbb{R}}$ and $t \in \mathbb{R}$ with the normalization that $\lambda((0, 1]) = 1$.

Note

For the *existence* of λ, will invoke an extension theorem of Caratheodory (to be done later)

Note

Interesting fact: there is no positive measure $\mu : 2^{\mathbb{R}} \to [0, \infty]$ which is translation invariant and $\mu((0, 1]) = 1$. This shows, in particular that $\mathcal{B}_{\mathbb{R}} \neq 2^{\mathbb{R}}$

3 Measurable functions

Definition 3.1 **Measurable function**

Let $(X, \mathfrak{M}), (Y, \mathfrak{N})$ be measurable spaces and $f : X \to Y$. Then f is called $\mathfrak{M}/\mathfrak{N}$-*measurable* if $f^{-1}(N) \in \mathfrak{M}$ for every $N \in \mathfrak{N}$.

Remark 3.2

Pre-images behave nicely with respect to the usual set operations.

Proposition 3.3 **Composition of measurable maps**

Compositions of measurable maps are measurable.

Proof: Obvious □

Proposition 3.4 **Measurability criterion**

Let $(X, \mathfrak{M})$, $(Y, \mathfrak{N})$ be measurable spaces. $f : X \to Y$, and $\mathfrak{C} \subseteq \mathfrak{N}$ such that $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{N}$ and such $f^{-1}(C) \in \mathfrak{M}$ for every $C \in \mathfrak{C}$. Then f is measurable.

Proof: Let $\mathfrak{Q} = \{Q \subseteq Y \mid f^{-1}(Q) \in \mathfrak{M}\} \subseteq 2^Y$

<u>*Claim:*</u> **$\mathfrak{Q}$ is a sigma algebra**

Note by assumption $\mathfrak{C} \subseteq \mathfrak{Q}$ thus $\mathfrak{N} = \sigma\text{-Alg}(\mathfrak{C}) \subseteq \mathfrak{Q}$. Thus f is measurable. □

Corollary 3.5 **Continuous maps are measurable**

Let X, Y be topological spaces, $f : X \to Y$ continuous. Then f is $\mathcal{B}_X/\mathcal{B}_Y$-measurable.

3.1 The space of functions $\mathrm{Bor}(X, \mathbb{R})$

Definition 3.6 **Borel measurable real-valued functions**

Given a measurable space $(X, \mathfrak{M})$. Define

$$\mathrm{Bor}(X, \mathbb{R}) := \{f : X \to \mathbb{R} \mid f \text{ is } \mathfrak{M}/\mathcal{B}_{\mathbb{R}} - \text{measurable}\}$$

Proposition 3.7 **Algebra and lattice operations**

Let $(X, \mathfrak{M})$ be a measurable space with $f, g \in \operatorname{Bor}(X, \mathbb{R})$. Then

1. $f + g \in \operatorname{Bor}(X, \mathbb{R})$
2. For any $c \in \mathbb{R}$, $cf \in \operatorname{Bor}(X, \mathbb{R})$
3. $f \cdot g \in \operatorname{Bor}(X, \mathbb{R})$
4. $f \vee g, f \wedge g \in \operatorname{Bor}(X, \mathbb{R})$

Proof of 3: Let $f, g \in \operatorname{Bor}(X, \mathbb{R})$. Let $h : X \to \mathbb{R}^2$ given by $h(x) = (f(x), g(x))$. Let $p : \mathbb{R}^2 \to \mathbb{R}$ given by $p(s, t) = s \cdot t$.

Claim: **h is measurable**

Claim: **p is measurable**

Claim: **$f \cdot g = p \circ h$**

Thus $f \cdot g \in \operatorname{Bor}(X, \mathbb{R})$ □

Exercise 3.1

Let $\mathfrak{S}$ be the collection of open squares in $\mathbb{R}^2$. Then $\sigma\text{-Alg}(\mathfrak{S}) = \mathcal{B}_{\mathbb{R}^2}$.

Remark 3.8

Let $(X, \mathfrak{M})$ be a measurable space. Then $\operatorname{Bor}(X, \mathbb{R})$ is an *algebra of functions* and a *lattice of functions*.

Remark 3.9

Operations with only one function. If $f \in \operatorname{Bor}(X, \mathbb{R})$, then $|f| \in \operatorname{Bor}(X, \mathbb{R})$. Use algebraic stability properties to prove this.

Example 3.1 **Problem 3.1**

Claim: **Let $\mathfrak{C} = \{(-\infty, t], \mid t \in \mathbb{R}\} \subseteq \mathcal{B}_{\mathbb{R}}$. Then $\sigma\text{-Alg}(\mathfrak{C}) = \mathcal{B}_{\mathbb{R}}$.**

Proof: Denote $\sigma\text{-Alg}(\mathfrak{C}) = \mathfrak{P}$. Clearly, $\mathfrak{P} \subseteq \mathcal{B}_{\mathbb{R}}$. If we can show that $G \in \mathfrak{P}$ for all $G \subseteq \mathbb{R}$ open, then we will be done because we will have that $\mathfrak{P}$ is a sigma-algebra and that $\mathfrak{P} \supseteq \mathcal{T}_{\mathbb{R}}$.
Now, every $\varnothing \neq G \subseteq \mathbb{R}$ open can be written as a countable union of open intervals. So then, it suffices to check that $\mathfrak{P}$ contains all of the open intervals of $\mathbb{R}$. This is left as an exercise. □

Let $(X, \mathfrak{M})$ be a measurable space, and suppose $f : X \to \mathbb{R}$ satisfies $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. That is, $f^{-1}(C) \in \mathfrak{M}$ for all $C \in \mathfrak{C}$. Then by Proposition 3.4, $f^{-1}(B) \in \mathfrak{M}$ for all $B \in \mathcal{B}_{\mathbb{R}}$, hence $f \in \operatorname{Bor}(X, \mathbb{R})$.

4 Convergence properties of $\mathrm{Bor}(X, \mathbb{R})$

Proposition 4.1 **Closure under suprema and infima**

Let $(X, \mathfrak{M})$ be a measurable space.

1. Let $f : X \to \mathbb{R}$ be a function. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that
$$\sup\{f_n(x) \mid n \in \mathbb{N}\} = f(x)$$
for all $x \in X$. Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$.
2. Let $g : X \to \mathbb{R}$ be a function. Let $(g_n)_{n=1}^{\infty}$ be a sequence of functions in $\mathrm{Bor}(X, \mathbb{R})$ such that
$$\inf\{g_n(x) \mid n \in \mathbb{N}\} = g(x)$$
for all $x \in X$. Then it follows that $g \in \mathrm{Bor}(X, \mathbb{R})$.

Proof:

1. It suffices to show that $f^{-1}((-\infty, t]) \in \mathfrak{M}$ for all $t \in \mathbb{R}$. This is left as an exercise.
2. Exercise.

$\square$

Remark 4.2

Let $(t_n)_{n=1}^{\infty}$ be a bounded sequence in $\mathbb{R}$. Then there exists $\ell_+ \in \mathbb{R}$ such that

1. There exists a subsequence of $(t_n)_{n=1}^{\infty}$ which converges to ℓ_+.
2. There is no subsequence of $(t_n)_{n=1}^{\infty}$ which converges to an $\ell > \ell_+$.

Define
$$\limsup_{n\to\infty} t_n := \ell_+$$
We can obtain ℓ_+ as
$$\ell_+ = \lim_{k\to\infty}\left(\sup_{n\geq k} t_n\right) = \inf_{k\geq 1}\left(\sup_{n\geq k} t_n\right)$$
We can define $\liminf_{n\to\infty} t_n$ similarly.

Fix a measurable space $(X, \mathfrak{M})$

Proposition 4.3

1. Let $f : X \to \mathbb{R}$. Suppose we could find $f_1, f_2, \ldots \in \mathrm{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(f_n(x))_{n=1}^{\infty}$ is bounded with

$$\limsup_{n \to \infty} f_n(x) = f(x)$$

Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$

2. Let $g : X \to \mathbb{R}$. Suppose we could find $g_1, g_2, \ldots \in \mathrm{Bor}(X, \mathbb{R})$ such that for every $x \in X$, the sequence $(g_n(x))_{n=1}^{\infty}$ is bounded with

$$\liminf_{n \to \infty} g_n(x) = g(x)$$

Then it follows that $g \in \mathrm{Bor}(X, \mathbb{R})$

Proof of 1: For every $k \in \mathbb{N}$, define the helper function $h_k : X \to \mathbb{R}$ by

$$h_k(x) = \sup \underbrace{\{f_n(x) \mid n \geq k\}}_{\text{bounded by hypothesis}}, \ \forall x \in X$$

Observe $f_k, f_{k+1}, \ldots \in \mathrm{Bor}(X, \mathbb{R})$. We can invoke Proposition 4.1, to conclude that $h_k \in \mathrm{Bor}(X, \mathbb{R})$. Observe for every $x \in X$, we have

$$h_1(x) \geq h_2(x) \geq \cdots$$

with

$$f(x) = \lim_{k \to \infty} h_k(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\}$$

This is because $f(x) = \limsup_{n \to \infty} f_n(x)$ by hypothesis. But then

$$h_k \in \mathrm{Bor}(X, \mathbb{R}), \ \forall k \in \mathbb{N}$$

and

$$f(x) = \inf\{h_k(x) \mid k \in \mathbb{N}\} \ \forall x \in X$$

thus $f \in \mathrm{Bor}(X, \mathbb{R})$ by Proposition 4.1. □

Proof of 2: We can reduce to (1), by introducing $f : X \to \mathbb{R}$, $f(x) = -g(x) \ \forall x \in X$ and introducing $f_n = -g_n \in \mathrm{Bor}(X, \mathbb{R})$, $\forall n \in \mathbb{N}$, then apply (1) to $f(x) = \limsup_{n \to \infty} f_n(x)$, to get $f \in \mathrm{Bor}(X, \mathbb{R})$, hence $g \in \mathrm{Bor}(X, \mathbb{R})$. □

Corollary 4.4

Let $f : X \to \mathbb{R}$. Suppose we could find $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}(X, \mathbb{R})$ such that $f(x) = \lim_{n \to \infty} f_n(x), \ \forall x \in X$. Then it follows that $f \in \mathrm{Bor}(X, \mathbb{R})$.

Proof: Use Proposition 4.3. □

Exercise 4.1

Suppose that $(X, \mathfrak{M}) = (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and consider the function $f' : \mathbb{R} \to \mathbb{R}$ (f' may not be continuous).

Solution:
Idea is to produce $(f_n)_{n=1}^{\infty}$ in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$ such that $\lim_{n\to\infty} f_n(x) = f'(x)$. If we find the f_n's then $f' \in \mathrm{Bor}(\mathbb{R}, \mathbb{R})$ due to Corollary 4.4. How do we find the f_n's? For $x \in \mathbb{R}$

$$f'(x) = \lim_{n\to\infty} \frac{f\left(x + \frac{1}{n}\right) - f(x)}{\frac{1}{n}} = \lim_{n\to\infty} n(u_n(x) - f(x))$$

With $u_n(x) = f\left(x + \frac{1}{n}\right)$. So then, for every $n \in \mathbb{N}$, let $u_n : \mathbb{R} \to \mathbb{R}$ be defined by $u_n(x) = f\left(x + \frac{1}{n}\right),\ \forall x \in \mathbb{R}$ and let $f_n = n(u_n - f)$. Every f_n is continuous, hence in $\mathrm{Bor}(\mathbb{R}, \mathbb{R})$ by Corollary 3.5 and $\lim_{n\to\infty} f_n(x) = f'(x)$ as we wanted.

4.1 Approximation by simple functions

Definition 4.5 **Simple functions**

A function $f \in \mathrm{Bor}(X, \mathbb{R})$ is said to be *simple* when it takes finitely many values, that is $f(X)$ is a finite subset of $\mathbb{R}$. We denote

$$\mathrm{Bor}_s(X, \mathbb{R}) := \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f \text{ is simple}\}$$

We also introduce the spaces

$$\mathrm{Bor}^+(X, \mathbb{R}) = \{f \in \mathrm{Bor}(X, \mathbb{R}) \mid f(x) \geq 0\ \forall x \in X\}$$

and

$$\mathrm{Bor}_s^+(X, \mathbb{R}) := \mathrm{Bor}_s(X, \mathbb{R}) \cap \mathrm{Bor}^+(X, \mathbb{R})$$

Note

For $A \in \mathfrak{M}$, we have $\chi_A \in \mathrm{Bor}_s(X, \mathbb{R})$, where $\chi_A : X \to \mathbb{R}$ is defined by

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \in X \setminus A \end{cases}$$

called the *indicator function of* A. Also note that χ_A is $\mathfrak{M}/\mathcal{B}_{\mathbb{R}}$-measurable since for every $S \in \mathcal{B}_{\mathbb{R}}$ we have that

$$\chi_A^{-1}(S) \in \{\varnothing, A, X \setminus A, X\} \subseteq \mathfrak{M}$$

Note

It can be shown that

$$\mathrm{Bor}_s(X,\mathbb{R}) = \underbrace{\mathrm{span}\{\chi_A \mid A \in \mathfrak{M}\}}_{\substack{\text{set of all linear combinations}\\ \text{of indicator functions}\\ \chi_A \text{ with } A\in\mathfrak{M}}}$$

Remark 4.6 **Binning values of a non-negative Borel function**

Let $f \in \mathrm{Bor}^+(X,\mathbb{R})$. For every $n \in \mathbb{N}$, let $f_n : X \to \mathbb{R}$ be the *n-th binning of the values of f*, defined as follows. For each $x \in X$:

1. If $f(x) \geq n$, define $f_n(x) := n$.
2. If $f(x) < n$, pick the unique $k \in \{1, \ldots, n \cdot 2^n\}$ such that

$$\frac{k-1}{2^n} \leq f(x) < \frac{k}{2^n}$$

   and define $f_n(x) := \frac{k-1}{2^n}$.

Proposition 4.7 **Binning approximation**

Let $f \in \mathrm{Bor}^+(X,\mathbb{R})$, and for every $n \in \mathbb{N}$, let f_n be the n-th binning of the values of f. Then

1. $f_n \in \mathrm{Bor}_s^+(X,\mathbb{R})$ for every $n \in \mathbb{N}$
2. $f_n \leq f_{n+1}$ for every $n \in \mathbb{N}$
3. For every $x \in X$,

$$\lim_{n\to\infty} f_n(x) = f(x)$$

5 Integration of functions in $\mathrm{Bor}^+(X,\mathbb{R})$

5.1 The map $L_s^+ : \mathrm{Bor}_s^+(X,\mathbb{R}) \longrightarrow [0,\infty]$

Fix a measure space $(X,\mathfrak{M},\mu)$

Note

For didactical reasons, it is convenient to first define $\int_X f(x)\,\mathrm{d}\mu(x)$ for $f \in \mathrm{Bor}^+(X,\mathbb{R})$. In this case, the integral is always defined but may be ∞. For general $f \in \mathrm{Bor}(X,\mathbb{R})$, the integral may not be defined; to get a finite integral, we need $f \in \mathcal{L}^1(\mu)$.
Moreover, it will be convenient to introduce the integral on $\mathrm{Bor}^+(X,\mathbb{R})$ as a map

$$L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \longrightarrow [0,\infty].$$

We will get L^+ in two parts:

1. Define $L_s^+ : \mathrm{Bor}_s^+(X,\mathbb{R}) \to [0,\infty]$
2. Extend L_s^+ to a map $L^+ : \mathrm{Bor}^+(X,\mathbb{R}) \to [0,\infty]$

Remark 5.1 **Easily verified properties of $\mathrm{Bor}_s^+(X, \mathbb{R})$**

1. If $f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$, then $f + g \in \mathrm{Bor}_s^+(X, \mathbb{R})$
2. If $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$, then $\alpha f \in \mathrm{Bor}_s^+(X, \mathbb{R})$
3. If $A \in \mathfrak{M}$, then $\chi_A \in \mathrm{Bor}_s^+(X, \mathbb{R})$

Note

As a consequence of (3), taking $A = \varnothing$ and $A = X$, we get $\underline{1}, \underline{0} : X \to \mathbb{R} \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

Theorem 5.2 **Integral of non-negative simple functions**

There exists a map

$$L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty],$$

uniquely determined, such that the following conditions are satisfied:

(i) Additivity:

$$L_s^+(f + g) = L_s^+(f) + L_s^+(g)$$

for all $f, g \in \mathrm{Bor}_s^+(X, \mathbb{R})$

(ii) Positive homogeneity:

$$L_s^+(\alpha f) = \alpha L_s^+(f)$$

for all $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ and $\alpha \in [0, \infty)$

(iii) Relation to the measure:

$$L_s^+(\chi_A) = \mu(A)$$

for all $A \in \mathfrak{M}$

In order to prove the existence of the map L_s^+, we will take advantage of the fact that one can describe the functions from $\mathrm{Bor}_s^+(X, \mathbb{R})$ in terms of measurable divisions of X.

Definition 5.3 **Measurable division and intersections of measurable divisions**

1. A *measurable division* of X is a set $\{A_1, \ldots, A_n\} \subseteq \mathfrak{M}$ of pairwise disjoint, non-empty sets such that $\bigcup_{i=1}^n A_i = X$
2. Given two measurable divisions $\Delta = \{A_1, \ldots, A_n\}$ and $\Gamma = \{B_1, \ldots, B_m\}$ of X, one can "intersect" them, to form a new measurable division of X, denoted $\Delta \wedge \Gamma$, defined by

$$\Delta \wedge \Gamma := \left\{A_i \cap B_j \;\middle|\; 1 \leq i \leq n, 1 \leq j \leq m, \text{ and } A_i \cap B_j \neq \varnothing\right\}$$

Lemma 5.4 **Simple functions and measurable divisions**

Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$.

1. There exists a measurable division $\Delta = \{A_1, \ldots, A_n\}$ of X and numbers $\alpha_1, \ldots, \alpha_n \in [0, \infty)$ such that

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$

2. Suppose that $\Delta = \{A_1, \ldots, A_n\}$ and $\alpha_1, \ldots, \alpha_n \in [0, \infty)$ are as in (1.), and suppose that $\Gamma = \{B_1, \ldots, B_m\}$ is a second measurable division of X. Then

$$f = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \alpha_i \chi_{A_i \cap B_j}$$

Proof of 2: For every $1 \leq i \leq n$, write

$$\chi_{A_i} = \chi_{A_i} \cdot 1 = \chi_{A_i} \left(\chi_{B_1} + \cdots + \chi_{B_m}\right) = \sum_{j=1}^{m} \chi_{A_i \cap B_j} = \sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \chi_{A_i \cap B_j}$$

Then

$$f = \sum_{i=1}^{n} \alpha_i \chi_{A_i} = \sum_{i=1}^{n} \alpha_i \left(\sum_{\substack{1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \chi_{A_i \cap B_j} \right) = \sum_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m, \\ A_i \cap B_j \neq \varnothing}} \alpha_i \chi_{A_i \cap B_j}$$

as desired. □

Lemma 5.5 **Well-definedness of the simple integral formula**

Let $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$. Suppose we have two ways of writing f as a linear combination:

$$f = \alpha_1 \chi_{A_1} + \cdots + \alpha_n \chi_{A_n}$$

and

$$f = \beta_1 \chi_{B_1} + \cdots + \beta_m \chi_{B_m},$$

where $\Delta = \{A_1, \ldots, A_n\}$ and $\Gamma = \{B_1, \ldots, B_m\}$ are measurable divisions of X, and $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m \in [0, \infty)$. Then

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{j=1}^{m} \beta_j \mu(B_j) \in [0, \infty].$$

Proof: For every $1 \leq i \leq n$, let us write

$$A_i = A_i \cap X = A_i \cap (B_1 \cup \dots \cup B_m) = \bigcup_{j=1}^{m} (A_i \cap B_j) = \bigcup_{\substack{1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} (A_i \cap B_j)$$

Since the union is disjoint,

$$\mu(A_i) = \sum_{\substack{1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \mu(A_i \cap B_j)$$

so

$$\sum_{i=1}^{n} \alpha_i \mu(A_i) = \sum_{i=1}^{n} \alpha_i \sum_{\substack{1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \mu(A_i \cap B_j) = \sum_{\substack{1 \le i \le n, \\ 1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \alpha_i \mu(A_i \cap B_j)$$

Similarly,

$$\sum_{j=1}^{m} \beta_j \mu(B_j) = \sum_{j=1}^{m} \beta_j \sum_{\substack{1 \le i \le n, \\ B_j \cap A_i \ne \varnothing}} \mu(B_j \cap A_i) = \sum_{\substack{1 \le i \le n, \\ 1 \le j \le m, \\ A_i \cap B_j \ne \varnothing}} \beta_j \mu(A_i \cap B_j)$$

Claim: **If $A_i \cap B_j \ne \varnothing$, then $\alpha_i = \beta_j$**

Observe that this claim will finish the proof. To verify the claim, suppose $A_i \cap B_j \ne \varnothing$. Pick $x \in A_i \cap B_j$, and evaluate $f(x)$ in two ways:

$$f(x) = \alpha_1 \chi_{A_1}(x) + \dots + \alpha_n \chi_{A_n}(x) = \alpha_i$$

similarly

$$f(x) = \beta_1 \chi_{B_1}(x) + \dots + \beta_m \chi_{B_m}(x) = \beta_j,$$

so $\alpha_i = f(x) = \beta_j$ ☐

Proof of existence of L_s^+ from Theorem 5.2: For every $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$, we define the quantity $L_s^+(f) \in [0, \infty]$ by the following procedure:

Consider a writing

$$f = \alpha_1 \chi_{A_1} + \dots + \alpha_n \chi_{A_n}$$

with $\Delta = \{A_1, \dots, A_n\}$ a measurable division of X and $\alpha_1, \dots, \alpha_n \in [0, \infty)$, (∗)
and put

$$L_s^+(f) := \sum_{i=1}^{n} \alpha_i \mu(A_i) \in [0, \infty].$$

Lemma 5.5 assures us that the procedure (∗) is coherent, in the sense that the proposed quantity $L_s^+(f)$ only depends on f and not on the choice of how we write f as a linear combination of indicators.

With $(*)$, we have thus defined a map

$$L_s^+ : \mathrm{Bor}_s^+(X, \mathbb{R}) \longrightarrow [0, \infty].$$

We must now check that this map has the properties (i), (ii), (iii) indicated in Theorem 5.2. □

5.2 The map $L^+ : \mathrm{Bor}^+(X, \mathbb{R}) \longrightarrow [0, \infty]$

Definition 5.6 **Integral of non-negative Borel functions**

For every function $u \in \mathrm{Bor}^+(X, \mathbb{R})$, we define

$$L^+(u) := \sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\} \in [0, \infty].$$

Remark 5.7 **Comments on the formula for L^+**

In the formula above, $f \leq u$ means $f(x) \leq u(x)$ for every $x \in X$. The set over which we take the supremum is non-empty because it contains $\underline{0}$. The supremum is taken in $[0, \infty]$, so it is allowed to be ∞.

Proposition 5.8 **L^+ extends L_s^+**

L^+ is an extension of L_s^+. That is, if $u \in \mathrm{Bor}_s^+(X, \mathbb{R})$, then the newly defined quantity $L^+(u)$ is the same as $L_s^+(u)$.

Proof: Must show

$$\sup\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\} = L_s^+(u)$$

"$\geq$" holds because L_s^+ is an element of $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. For "$\leq$", we must show $L_s^+(u)$ is an upper bound for $\{L_s^+(f) \mid f \in \mathrm{Bor}_s^+(X, \mathbb{R}), f \leq u\}$. Indeed, if $f \in \mathrm{Bor}_s^+(X, \mathbb{R})$ with $f \leq u$, then, using Theorem 5.2,

$$L_s^+(u) = L_s^+(f + (u - f)) = L_s^+(f) + L_s^+(u - f) \geq L_s^+(f)$$

□